

Annotating single-cell data on a laptop: a 13-method benchmark and practical low-memory workflows, with actinn-jax

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Abstract

Reference-based cell-type annotation is a routine, high-volume step in single-cell analysis, and most labs run it on a laptop, not a GPU cluster. We present **actinn-jax**, a from-scratch JAX reimplementation of ACTINN (a 4-layer MLP classifier) with a sparse-aware pipeline and a train-once/map-many cached reference model, and use it as the occasion for a neutral benchmark of **thirteen** methods spanning every major class — classical (SVM, kNN), regularized-linear (CellTypist), a carefully-tuned linear pipeline, parameter-free projection (scTOP), marker/correlation (SingleR, scmap), deep probabilistic (scANVI, scArches), an interpretable prototype VAE (ProtoCloud), and a foundation model (scPRINT) — across **six** datasets (within-dataset, cross-dataset and cross-study; 8 to 86 cell types) plus a five-point **scaling sweep to 49k reference cells**, measuring accuracy, macro-F1, ontology-aware concordance, training and inference time, and peak memory on commodity Apple-Silicon hardware. **Accuracy differences among the leading methods are small; their cost differences are not.** The top of the accuracy table is a four-way cluster spanning 0.008 (0.831–0.839), led by a carefully preprocessed **linear pipeline** (normalize → ANOVA → PCA → logistic regression, 0.839) rather than by a deep model, with scANVI, scArches and actinn-jax (0.831) alongside it. **Those same four methods differ by ~130× in inference time** (0.34 s to 89 s) **and 2.4× in peak memory** (1.8 to 4.3 GB). The linear pipeline is the most accurate-per-second option on laptop-sized data, fitting 7× faster than actinn-jax (3.0 vs. 21.9 s). At atlas scale the ordering changes: **ProtoCloud** rises from worst at 3k reference cells (0.722) to best at 49k (**0.976**, vs. 0.936 for actinn-jax and 0.939 for the linear pipeline). No method leads everywhere, and methods separated by less than a percentage point of accuracy are separated by two orders of magnitude in cost. Because several methods now annotate quickly, accurately, and with a usable handle on unknown cells, the useful contribution is not another ranking but an account of **which tool fits which job, and what the surrounding workflow looks like**. Every method is run on every dataset through one harness on identical splits, the panel deliberately includes the baselines most likely to beat a small MLP, and the external validation is somebody else’s benchmark — **Open Problems (OP) label_projection**, whose datasets, metrics and ranking we did not choose. We report the results as a decision aid (accuracy-per-second, accuracy-per-byte, and behaviour from 3k to 49k reference cells), then use actinn-jax to demonstrate **end-to-end workflows that a low, flat cost profile makes practical**, on human data and — via a Cell-Ontology-derived hierarchy that takes the accelerator out of the build — on mouse: annotating an unknown human dataset from a **shipped census-scale reference** (~800 cell types, calibrated abstain) with no training; **tissue-aware refinement** that halves spurious cross-tissue labels on a liver query; a **broad→refined hand-off** to a small focused reference (held-out cross-study liver 0.23/0.58 → 0.72/0.86, exact-match / Cell-Ontology-aware); within-cell-type resolution (hepatocyte zonation); and a **cluster-level novel-cell-type screen** that recovers a withheld pulmonary ionocyte population and its marker ASCL3. What enables this is inference that is **sub-second and flat** — independent of reference size and cardinality — over a cached reference that is trained once and reused, at the lowest peak memory of the methods we carried to atlas scale (6.1 vs. 13.2 GB for the linear pipeline at 49k cells). The components are not unique: ProtoCloud provides uncertainty, gene attribution and a retraining-

based refinement path, and **Pan-human Azimuth** ships a pretrained, calibrated, hierarchical pan-human annotator that runs on a laptop — a better-resourced broad tier than ours, and one we can **distill into a broad-pass model of comparable accuracy at $\sim 9\times$ its throughput, built with no GPU and no labeled data**. What the workflow adds is the **hand-off into a label set the broad model was never trained on**: refinement against a user’s own focused reference, and resolution below any fixed typology’s leaves.

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1 Introduction

Annotating cell types by mapping to a labeled reference is one of the most frequently run operations in single-cell RNA-seq. The method landscape is large (survey), but published comparisons ([Abdelaal 2019], [Huang 2024]) emphasize accuracy and rarely report the axis that decides what a working scientist actually runs on their own machine: **wall-clock time and memory on commodity hardware**, without a GPU. Foundation models (scGPT, Geneformer, scPRINT) push accuracy in some settings but need GPUs and minutes-to- hours, and — as we and others find — their *zero-shot* label predictions underperform a small model trained on curated reference labels.

We ask a practical question: **for a given annotation job on commodity hardware, which method should you actually run, and what does the surrounding workflow look like?** Several methods now annotate quickly and accurately and give a usable signal on unknown cells, so the shortage is not another leaderboard but guidance on fit-for-purpose — accuracy per second, accuracy per byte, and how each behaves as the reference grows. We answer with a neutral 13-method benchmark plus a set of demonstrated workflows, held to a single standard: every method runs in its own environment through the same harness, on the same splits, scored by the same metrics, and reported as measured — including where *actinn-jax* is not the leader.

How the comparison was constrained. A benchmark written by a method’s own author has a well-known failure mode: the datasets, baselines and metrics that flatter the method are the ones that get reported. We tried to design that freedom out rather than promise restraint. Every method runs on **every** dataset, so none appears on a favourable subset, and the two methods outside that matrix are named with reasons (§2.2). The panel was chosen to include the baselines most likely to beat a small MLP — a carefully tuned linear pipeline, scTOP, ProtoCloud — and each of them does beat it somewhere; our own classical tier is left untuned while the linear baseline is tuned, which is the unfavourable direction (§5). Metrics are fixed across all methods and reported in full. Most importantly, the **external validation is somebody else’s**: Open Problems `label_projection` (§3.7) sets the datasets, the metrics and the ranking, and we did not choose any of them. Where results contradict our own earlier claims or our method’s interests — ProtoCloud overtaking *actinn-jax* at atlas scale, a bounded rather than widening memory advantage, a better-resourced broad annotator than our own — they are reported in those terms.

On adding a twelfth method to a crowded field. We are adding one, and we think it earns its place: the original ACTINN no longer installs or runs on current toolchains, and the reimplementations’ flat, sub-second, memory-bounded inference over a cached reference is what makes the multi-stage workflow of §3.4 practical on a laptop. What we are *not* doing is the move that makes method papers hard to trust — concluding that the new method wins. It does not: on accuracy it sits inside a four-way cluster it does not lead, the panel was assembled to include the baselines most likely to beat it, and they do. A field with a dozen working methods ([xkcd 927]) is

a good reason to be specific about what a new entry is *for* — here, a cost profile that lets several models be chained — rather than to claim it supersedes what exists. Where a better-resourced model already exists, the productive move is to build on it (§3.4).

Contributions. 1. A modern, dependency-light (no TensorFlow) JAX reimplement of ACTINN with sparse preprocessing, chunked atlas-scale prediction, and a cached reference model. 2. A neutral **13-method** $\times$ **6-dataset** benchmark of accuracy, speed, and memory on Apple Silicon, with a rejection/abstain analysis and a **five-point scaling sweep to 49k reference cells**. The panel deliberately includes the baselines most likely to beat a small MLP — a tuned linear pipeline, scTOP, and ProtoCloud — and the results are reported as measured (§3.1, §5). The sweep is also what sets actinn-jax’s minibatch policy: at ACTINN’s fixed batch of 128, a 47k-cell reference costs $\sim 24k$ tiny update steps, so actinn-jax scales the batch with the reference above $\sim 12.8k$ cells and keeps 128 below it. Measured fit and predict curves are in §3.3. 3. **Demonstrated end-to-end workflows** for jobs where cost is the binding constraint, all CPU: annotate an unknown human dataset from a **shipped census-wide ~ 800 -type reference** with a calibrated abstain and no training; **tissue-aware refinement** (halves spurious cross-tissue labels on a liver query); a **broad $\rightarrow$ refined hand-off** to a small focused reference (cross-study liver $0.23/0.58 \rightarrow 0.72/0.86$); within-cell-type resolution (hepatocyte zonation); and a **one-call novel-cell-type screen** (recovers a withheld pulmonary ionocyte population and its marker ASCL3). We are precise about what is not ours: ProtoCloud offers uncertainty, attribution and a retraining-based refine, and Pan-human Azimuth ships a pretrained hierarchical pan-human annotator with trained abstention, so the broad tier itself is not a distinguishing feature — we distill that model into a broad-pass reference rather than compete with it. What is distinct is **refinement into a label set no pretrained model carries** — the user’s own focused reference, and states below a fixed typology’s leaves. The same route builds a **pan-mouse reference** (453 types / 85 tissues; 0.638 ontology concordance on two withheld datasets) in 17 s of CPU, using Cell Ontology lineage in place of a foundation-model embedding — which beats the embedding-derived hierarchy on the human corpus it was validated against, and needs no GPU at any stage. 4. An **independent external validation** on Open Problems **label_projection**, with a controlled same-hardware speed/memory comparison, and a set of cheap ablations that characterize *when* the gene-space model gains or loses to heavier methods (input standardization; a reference-guided gene budget; a foundation-model negative control).

2 Methods

2.1 actinn-jax

actinn-jax reimplements ACTINN ([Ma & Pellegrini 2020]) — a 4-layer fully-connected network (100/50/25 hidden units, ReLU, softmax) trained with Adam — in JAX/optax ([Bradbury 2018]), replacing the original’s TensorFlow-1.x graph/session code. Key engineering: - **Sparse-aware preprocessing**: no `adata.to_df` densification; CP10k+log2 normalization and the expression / coefficient-of-variation gene filter run on sparse matrices; only the selected-gene columns of each minibatch are densified. - **Cached reference model** (ReferenceModel): train once, `save/load`, map many queries — the amortized cost of repeated annotation against a fixed reference drops to the inference cost alone. - **Chunked prediction** for atlas-scale queries; `use_raw='auto'` to pick raw counts from `.raw` (CELLxGENE convention). - **Optional reference-fit standardization** (`standardize=True`): z-score each selected gene by the reference’s frozen mean/std and apply it to the query — a cheap domain alignment (§3.7). Opt-in, with a saved scaler that projects with the model; μ/σ persist in the `.npz` (old models load standardization-off, backward-compatible). The

reimplementation reproduces the original’s accuracy to within repeat noise while being **substantially faster and lighter and, unlike the TensorFlow-1.x original, installable on current Python/ML environments** — the original’s `tf.compat.v1` graph/session code no longer runs on a modern stack without pinning a years-old TensorFlow. Because the two are the same model at equal accuracy, we do not carry the original as a separate benchmark row; the engineering win is the point, not an accuracy comparison.

2.2 Benchmarked methods

method	source	tier	engine	rejection	env
actinn-jax	[Ma & Pellegrini 2020]	classical	JAX MLP	abstain (min_prob)	core
SVM	[Pedregosa 2011]	classical	linear SVM (SGD)	—	core
kNN	[Pedregosa 2011]	classical	k-nearest neighbors	—	core
CellTypist	[Domínguez Conde 2022]	linear	L2 logistic regression	prob threshold	core
linear-anova-pca	[Pedregosa 2011] †	linear	normalize→ANOVA→PCA(220)-edoreg	—	—
scTOP	[Yampolskaya 2023]	parameter-free	rank z-score class-average projection	—	core (sctop)
SingleR	[Aran 2019]	correlation	Spearman + fine-tuning	—	R/.Rlib
scmap-cluster	[Kiselev 2018]	correlation	centroid cosine	yes (unassigned)	R/.Rlib
scANVI	[Xu 2021]	deep	scVI semi-supervised VAE	prob	.venv-scvi (MPS)
scArches	[Lotfollahi 2022]	deep	scANVI reference surgery	prob	.venv-scvi (MPS)
ProtoCloud	[Guo & Ding 2026]	deep	prototype VAE + LRP attribution	ambiguity flag	.venv- protocloud
scPRINT	[Kalfon 2025]	foundation	pretrained transformer, zero-shot	—	.venv-scprint (MPS)
Pan-human Azimuth	[Sarkar et al. 2026]	pretrained	8-level hierarchical NN, fixed 382-leaf typology	trained Unassigned	.venv- panhuman

† **linear-anova-pca** has no method paper: it is a baseline assembled here from scikit-learn com-

ponents, tuned deliberately to be the strongest simple competitor we could build (§1).

Eleven of the thirteen are trained on each dataset’s own reference and run the full accuracy/cost matrix of §3.1 on all six datasets, so none is compared on a favourable subset. The remaining two are **pretrained annotators with fixed label vocabularies**, which changes what can be measured rather than excusing them from measurement:

- **scPRINT** is a zero-shot foundation model over a fixed **Cell Ontology (CL)** vocabulary — CL being the standard controlled vocabulary of cell types, used throughout as the common ground between datasets that name the same cell differently; it cannot emit labels outside that set and skips symbol-keyed datasets.
- **Pan-human Azimuth** ([Sarkar et al. 2026]) is a supervised hierarchical classifier over a single organism-wide typology — 8 levels, 382 leaves, ~7M parameters, a fixed 5,055-gene panel, trained on 9.7M curated cells, with abstention *learned rather than thresholded* — the model is trained to answer **Unassigned**, instead of having a probability cutoff applied to its output afterwards. It is the closest published counterpart to this paper’s broad pass, and the teacher we distill in §3.4. Adapter: [panhuman_adapter.py](#).

Neither predicts into the dataset’s label strings, so **exact-match accuracy against those strings is a vocabulary artifact, not an accuracy signal** — regulatory T cell → Treg cell and non-classical monocyte → CD16 monocyte are correct answers that score zero. Both are therefore scored **ontology-only** (§2.4) and reported in their own block in §3.1, on the three datasets where reference and query both carry Cell Ontology ids. They appear again in §3.6 (foundation-model zero-shot) and §3.4 (Azimuth as broad pass, and as distillation teacher).

scPred is excluded outright: unmaintained; it requires harmony’s removed **HarmonyMatrix** API and older harmony versions do not compile on the current toolchain.

2.3 Datasets

dataset	source	split	tissue	#types	genes	notes
lung_intra	[Travaglini 2020]	within-dataset	lung (Krasnow)	46	Ensembl	300 cells/type ref
lung_cross	[Sikkema 2023] → [Travaglini 2020]	cross-dataset	lung (HLCA→Krasnow)	46	Ensembl	different lab/protocol
liver_intra	[Edgar 2026]	within-dataset	liver (HLiCA)	36	Ensembl	150 cells/type
liver_cross	[Edgar 2026]	cross-study	liver (HLiCA)	34	Ensembl	train 6 studies → test withheld study
blood_gut_intra	[Alegbe 2026]	within-dataset	blood + gut (IBDverse)	86	Ensembl	high cardinality; no CL ids

dataset	source	split	tissue	#types	genes	notes
pbmc	[10x Genomics 2016]	within-dataset	PBMC (pbmc3k)	8	symbols	small-n; scPRINT skips (symbols)

The blood+gut set is a subsample of **IBDverse** (Wellcome Sanger Institute; blood, terminal ileum and rectum from 421 individuals), included here because 86 fine-grained labels make it the high-cardinality stress case for the panel. It carries no Cell Ontology ids, so it contributes to accuracy and macro-F1 but to no ontology-scored result.

The set spans the three generalization regimes (within-dataset, across datasets, across studies), an order of magnitude in cell-type count (8→86), and both gene-ID conventions.

2.4 Metrics

Per (dataset, method, repeat): **accuracy**; **macro-F1**, unweighted over classes so rare types count as much as common ones; **ontology-aware concordance**; **fit time**, **predict time**, and **peak RSS** (resident set size — the maximum physical memory the process held).

Ontology-aware concordance credits a call that is the same node, an ancestor, or a descendant of the truth in the Cell Ontology. It exists because cell-type vocabularies disagree about granularity more than about biology: **perportal region hepatocyte** against **Hepatocyte** is a right-lineage-wrong-depth call, which exact match scores as a total failure and a working scientist would accept. It is the **only** metric that compares methods with different label vocabularies (§2.2), and it is reported only where both reference and query carry CL ids — so not on **blood_gut_intra**, and not on the symbol-keyed **pbmc**.

Concordance depends on the ontology release, since ancestor sets change between them. All numbers here use **cl-basic.obo**, **CL release 2026-06-08** (sha256 73996c63...), resolved once and reused by every scoring pass (**benchmark/metrics.py**, via **pronto**). This is not a hypothetical concern: re-running **lung_cross** against a later-fetched release reproduced the splits exactly (14,390 reference / 13,550 query cells) and still moved concordance by 0.003, with no model changing. The release, its checksum, and exact versions for all six benchmark environments are recorded in **envs/locks/** (**tools/freeze_envs.py --check** reports drift).

Exact-match accuracy is reported for the ten reference-trained methods and, on **lung_cross**, flagged: reference and query use different label vocabularies there, so its exact accuracy is a vocabulary artifact and the ranking, not the level, is what transfers.

2.5 Harness, isolation, and hardware

Each method runs in **its own virtual environment via subprocess isolation** (**benchmark/runner.py**), because the dependency sets are mutually unsatisfiable — scVI-based methods, TensorFlow/Keras (Pan-human Azimuth), R (SingleR, scmap) and the JAX core cannot coexist in one interpreter. The driver (**benchmark/driver.py**) builds the reference/query split **once per dataset**, from a fixed seed, and hands the identical pair to every method, so no method sees a different split. **repeats = 3**; scPRINT and Pan-human Azimuth run once each, being deterministic given a fixed query and dominated by a single forward pass.

Resource accounting is per subprocess: fit and predict are timed separately and peak RSS is sampled by a monitor thread in the child, so one method’s footprint cannot be attributed to another.

Hardware: Apple Silicon (M-series), CPU for the classical, linear, correlation and pretrained tiers and for actinn-jax; Apple MPS for the deep and foundation tiers. No discrete GPU, no cloud — the regime the benchmark is about. Math-library threads are left uncapped, so wall-clock is what a practitioner sees; an optional thread cap exists for stricter reproducibility. §3.7 repeats the comparison on one cloud CPU instance, where matched concurrency matters and is discussed (§2.10).

2.6 Building a broad reference

The shipped broad-pass references (§3.4) are built once, offline, by a three-stage pipeline driven by one command (`update_broad_reference.sh`;).

Sampling. All primary cells for one organism in the CELLxGENE Census [CZI Census 2025] (`is_primary_data == True`), stratified by `cell_type` at a fixed cap per type — 40 for `broad_human_v1`, 60 for the later human and mouse pulls — dropping types with fewer than 12 cells and the uninformative labels (`unknown`, `native cell`, `eukaryotic cell`, `animal cell`). The pull is checkpointed per batch of ten datasets and retried, since transient TileDB/S3 read errors are routine at this scale. Both `feature_id` (Ensembl) and `feature_name` (symbol) are carried, because symbol-keyed consumers cannot use an Ensembl-only pull. `stable` is a moving pointer, so the resolved release is recorded; every reference here comes from **census 2025-11-08**.

Coarse hierarchy — two interchangeable routes. The model is a coarse classifier over groups plus one fine classifier per group, and the grouping comes from either:

1. **Foundation-model embeddings.** `scPRINT (medium-v1.5)` embeds the reference once in 4,000-cell chunks; per-cell-type centroids are Ward-clustered into `max(8, round( $\sqrt{n\_types}$ ))` groups. `scPRINT`’s QC drops cells, so the embedding is not positionally aligned to the input and the label is carried through the model with the vectors. This is the only step that wants a GPU/MPS.
2. **Cell Ontology lineage.** Each cell type is described by a binary indicator over the CL terms it descends from; terms appearing in fewer than two types, or in all of them, are dropped as uninformative; the same Ward clustering is applied to those vectors (`ontology_hierarchy.py`). Free, deterministic, and species-independent — which is what makes the pan-mouse reference possible, since the distillation teacher is human-only and `scPRINT`’s mouse support is untested here. §3.4 compares the two routes against a random-grouping control.

Training and calibration. A 4,000-gene panel of **highly variable genes (HVGs)** — the genes that differ most across cells, and so carry most of the signal for telling types apart — is selected on the reference; the coarse and per-group fine models are trained on it. Abstain calibration holds out **10% of cell types entirely** as out-of-distribution (OOD — types the model was never trained on, standing in for the novel populations a real query contains), plus a 20% within-type test split, and sweeps `min_prob` to trade coverage against accuracy on kept cells — the table each shipped reference reports.

Shipping and verification. Each reference is written with a `build_info.json` recording census release, corpus sizes, hierarchy route, organism and calibration table. Because a rebuild can train cleanly and annotate *worse*, the previous reference is backed up and the new one is scored on a held-out atlas before it is kept (`verify_reference.py`). For `broad_mouse_v1` the test set is carved

from the census itself: two datasets are excluded from the reference sample and re-pulled as the query, so “held out” means held-out *datasets* and not merely held-out cells (EXCLUDE_DATASETS / ONLY_DATASETS).

2.7 Distilling a pretrained annotator

A broad reference can also be built without labels or an accelerator, by taking both the label vocabulary and the hierarchy from an existing pretrained annotator. Keras/TensorFlow and JAX cannot share a process, so this runs in two stages.

Teacher pass (`distill_dump.py`, `.venv-panhuman`). Pan-human Azimuth labels a corpus of raw counts through its low-level API (`AzimuthNN_base`, batch 8,192), which loads weights once rather than per call. All eight levels are retained. Where the package cannot reconcile a refined level with its coarser ones it blanks that column to boolean `False`; those cells still carry a deepest-level call, so we fall back to it rather than discard them — dropping them would distill only the cells the teacher found easy. Predictions are mapped to CL ids through the package’s own crosswalk.

Student pass (`distill_train.py`, `core` `venv`). The student is a `HierarchicalReferenceModel` whose `classes` are the teacher’s refined fine labels (including its trained `Unassigned` quality-control class) and whose `hierarchy` is the teacher’s own broad level — so the coarse→fine structure is inherited rather than rediscovered. 4,000-gene HVG panel; classes with fewer than 8 cells dropped, since a class that cannot be split into train and test makes fidelity depend on which side its cells landed.

Corpus. Three local atlases (lung, liver, blood+gut) capped at 600 cells per label, plus a census-wide pull (≤ 60 per type). The atlases’ own labels are used **only for scoring**, never for training.

Evaluation. Two arms answer different questions. *In-corpus* holds out a stratified 25% of cells and measures **fidelity** — how often the student reproduces the teacher, exactly and ontology-equivalently. *Held-out atlas* withholds an entire atlas and measures whether the student generalizes to data the corpus never covered. Both arms also score student and teacher against the atlases’ own labels, which separates distillation loss from teacher error. Note that once the census pull is in the corpus, a withheld *atlas* is no longer a withheld *tissue*.

2.8 Refinement, abstention, and novelty

Four mechanisms operate on a trained reference at inference time, all label-free.

- **min_prob abstention.** Cells whose final probability is below the threshold are relabeled `unknown` rather than forced to the nearest class. Calibrated per reference (§2.6).
- **refine_to_query.** The reference’s class set is masked to those classes the query’s own predictions actually support, and the softmax renormalized — no ground truth, no retraining.
- **refine_to_tissue.** Classes are masked to those the census records in the given tissue, while **pan-tissue** types (immune, endothelial, stromal) stay available everywhere, so liver-resident T cells are unaffected. The per-class `tissue_general` map is baked into the shipped reference; tissue is passed explicitly or read from `adata.obs`, with common synonyms (PBMC→blood, hepatic→liver) recognized. A tissue outside the reference’s vocabulary imposes no filter rather than an empty one.
- **Cluster-level novelty screen.** A cell type absent from the reference appears not as scattered uncertain cells but as a *coherent group the reference cannot confidently explain*. `detect_novel_celltypes` flags clusters that are both large enough to be a real population

(`min_cells`) and predominantly low-confidence, and reports marker genes for each. This is distinct from per-cell abstention.

Within-type state. The same machinery resolves structure below a cell-type label: a hepatocyte zonation model trained on portal→central zone labels, scored by within-one-zone agreement across donors and datasets.

2.9 Scaling protocol

The reference is subsampled to five sizes up to **49k cells** with the query held fixed, so that accuracy, fit time, predict time and peak memory are functions of reference size alone. Reported per method; the point of interest is which curves cross.

2.10 External validation protocol

Open Problems label_projection [Open Problems 2024] supplies the datasets, metrics and ranking — none chosen by us. `actinn-jax` is packaged as a `viash 0.9.7` component (`config.vsh.yaml` + `script.py`) declaring its `preferred_normalization`, and run through the project’s own Nextflow workflow. We additionally built components for four baselines (linear-anova-pca, scTOP, SVM, CellTypist) so the same-hardware comparison covers more than the leaderboard’s own set.

Two caveats govern how those numbers may be read. The leaderboard aggregates runs from **heterogeneous cloud hardware**, so its timings are not a controlled cost comparison — hence the same-hardware run: every method on one AWS CPU instance. And in that run, **wall-clock is only meaningful at matched concurrency**: with several tasks per box, %cpu spanned 49% to 2338% across methods, so cross-method runtime ratios measured at high concurrency rank threading and scheduler contention as much as algorithm. Where matched-concurrency data is missing, §3.7 reports coverage and memory and declines to report runtime ratios.

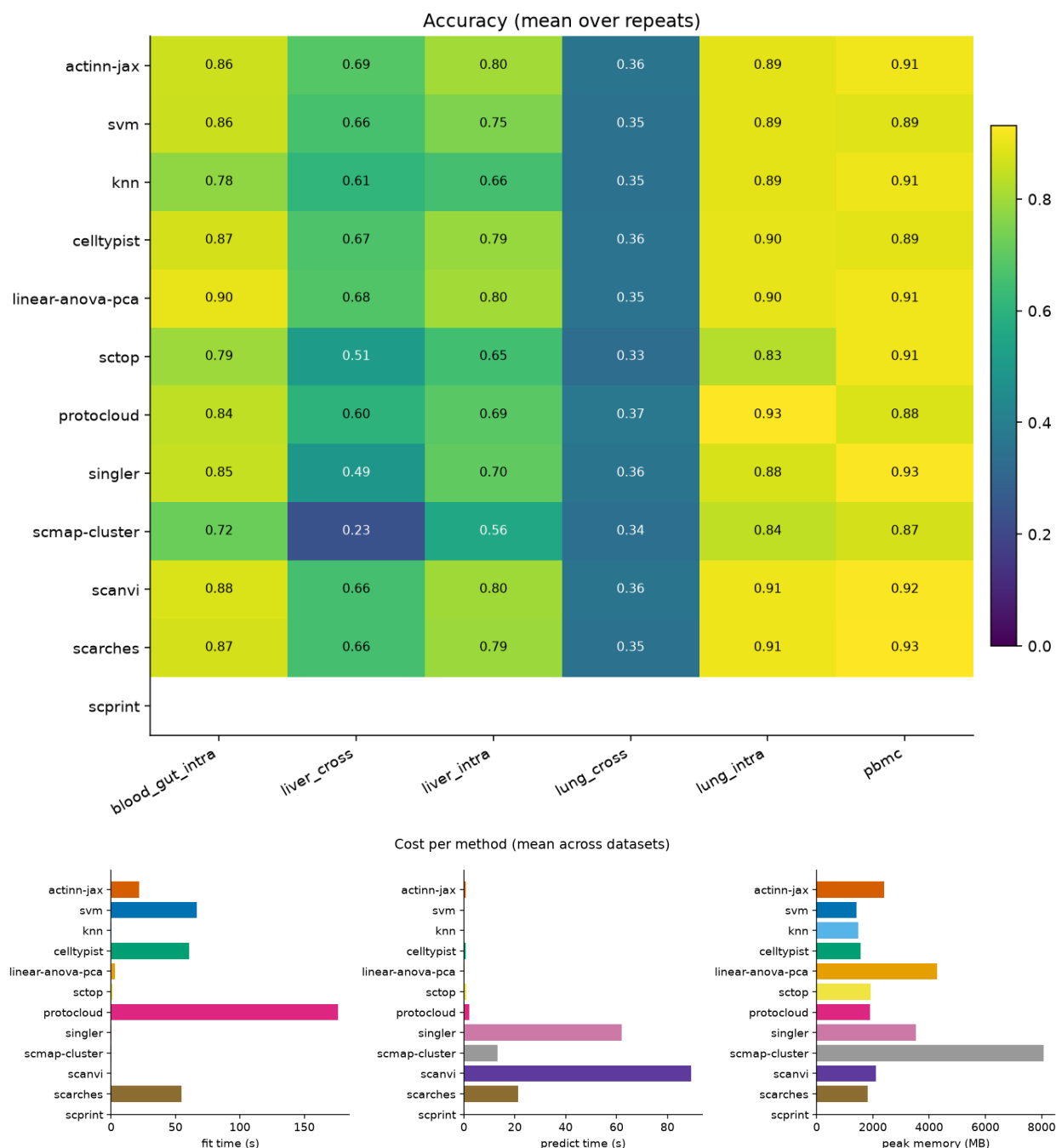
2.11 Ablations

Three cheap interventions characterize *when* a gene-space MLP gains or loses against heavier methods, all selectable without test labels:

- **Input standardization** — z-score each selected gene by the reference’s frozen mean/std and apply to the query (a scArches-style domain alignment). Opt-in, because it shifts the probability calibration the abstain thresholds are tuned against.
- **Gene budget** — Open Problems feeds every method 1,000 HVGs; we sweep wider panels and select per dataset by **held-out reference cross-validation**, which is label-free with respect to the test set.
- **Protein-embedding featurization** (negative control) — a CPU-only featurization in the style of **Universal Cell Embeddings (UCE)** [Rosen 2026] — an expression-weighted mean of ESM2 [Lin 2023] protein-language-model gene embeddings — to test whether a foundation model’s value survives a cheap pooling shortcut.

3 Results

3.1 Accuracy and cost



Accuracy and cost belong in one table: the finding is not where either lands, but how little they relate. Accuracy is the mean over the **five shared-vocabulary datasets** (lung_cross excluded — its exact accuracy is a vocabulary artifact, see the † note below; means over all six are ~0.08 lower for every method, with the identical ranking); macro-F1 and ontology are means over all six; cost is the mean per query. Ordered by accuracy:

method	acc (5)	macro-F1		fit (s)	predict (s)	peak MB
		(6)	ontology (6)			
linear-anova-pca	0.839	0.699	0.808	3.0	0.34	4294
scANVI	0.833	0.697	0.809	0.0*	89.2	2099
scArches	0.832	0.699	0.805	54.7	21.4	1819
actinn-jax	0.831	0.683	0.811	21.9	0.67	2399
CellTypist	0.823	0.691	0.805	60.7	0.66	1569
SVM	0.810	0.680	0.797	66.5	0.07	1415
ProtoCloud	0.790	0.649	0.778	176.3	2.14	1893
SingleR	0.770	0.652	0.750	0.3	62.1	3541
kNN	0.770	0.622	0.769	0.4	0.15	1483
scTOP	0.739	0.619	0.703	1.1	1.07	1928
scmap-cluster	0.646	0.550	0.771	0.4	13.1	8073

(*scANVI does most of its work in one train+predict pass, attributed to predict.)

Read down the accuracy column, then across. The top four span 0.008 in accuracy, $\sim 130\times$ in predict time (0.34 s to 89 s) and $2.4\times$ in memory. scANVI sits within 0.002 of second place and **134 $\times$ slower than actinn-jax** at tied accuracy. The linear pipeline is both the most accurate and the fastest to fit, and pays for it in memory — 4294 vs 2399 MB — because ANOVA/PCA densify a cells $\times$ genes matrix while actinn-jax stays sparse; that $\sim 1.8\times$ ratio widens to $\sim 2\times$ at atlas scale (6.1 vs 13.2 GB at 49k cells, §3.3). The two profiles suit different jobs: one-shot labelling, versus a reference **trained once and reused**, where a cached model pays fit once while the linear pipeline refits scaler/PCA/classifier for every query. ProtoCloud’s 146 s CPU fit buys nothing at this reference size — and buys the top accuracy at atlas scale (§3.3).

Pretrained annotators, scored ontology-only. The two methods with fixed label vocabularies (§2.2, §2.4) cannot appear in the table above, since they do not predict the dataset’s label strings. They are scored by ontology concordance on the three datasets where reference and query both carry CL ids, with actinn-jax on the same splits for reference:

	lung_intra	liver_intra	liver_cross	predict (s)
actinn-jax (reference-trained)	0.917	0.846	0.731	0.27–0.37
Pan-human Azimuth (pretrained)	0.700	0.521	0.408	2.2–3.4
scPRINT (zero-shot)	0.201	—	—	62.3

This is not a like-for-like comparison and should not be read as one. actinn-jax is trained on a reference drawn from the same data and scored in its own vocabulary; Pan-human Azimuth

has never seen these datasets and answers in a different one. A reference-trained model *should* win. What the block does establish is the distance between a curated supervised annotator and a zero-shot foundation head — 0.700 against 0.218 on lung — and it sets up §3.4, where Azimuth serves as a broad pass in its own right and as a distillation teacher. Peak memory is indistinguishable between actinn-jax and Azimuth (1.65–2.1 GB on every dataset).

The top of the accuracy table is a **four-way tie within 0.008** — the tuned **linear pipeline (0.839)**, scANVI (0.833), scArches (0.832) and actinn-jax (0.831) — led by the linear pipeline rather than by a deep model. CellTypist (0.823) sits just outside it. actinn-jax has the best ontology-aware concordance (0.811), marginally ahead of scANVI’s 0.809 — a gap well inside repeat noise, so read it as a tie rather than a lead. ProtoCloud (0.790) and scTOP (0.739) sit below that cluster on these subsampled references: both need conditions these splits do not provide — ProtoCloud is data-hungry and becomes the strongest method once given a real atlas (§5, `SCALING_MEMORY.md`), while scTOP is built for small, low-cardinality problems. Per dataset (accuracy):

dataset	actinn-jax	best method (acc)
lung_intra (46 types)	0.894	ProtoCloud 0.932
lung_cross (cross-dataset)†	0.358 exact / 0.752 ontology	ProtoCloud 0.374 / 0.791 ontology
liver_intra (36 types)	0.802	linear-anova-pca 0.804 (actinn-jax #2)
liver_cross (cross-study)	0.686 exact / 0.731 ontology — #1 on both	actinn-jax
blood_gut (86 types)	0.860	linear-anova-pca 0.902
pbmc (8 types)	0.913	scArches 0.931

No single method leads everywhere: the linear pipeline and ProtoCloud each take two datasets, actinn-jax and scArches one apiece. actinn-jax leads the cross-study liver split — the hardest generalization regime here, and the one closest to real reference mapping — and is second on within-dataset liver, while trailing on lung, on the 86-type blood+gut set (−4.2 pt to the linear pipeline), and on small-n pbmc. The spread across the leading methods on any one dataset is 1–4 points. scmap-cluster is consistently the weakest (0.23 macro-F1 on liver_cross).

† **lung_cross exact accuracy (~0.35 for every method) is a label-vocabulary artifact, not a transfer failure.** HLCA (reference) and Krasnow (query) were annotated independently and share only **20 of 46 cell-type names**; 26 of Krasnow’s types are absent from HLCA’s vocabulary, so no classifier can emit the exact string. The mismatch is granularity: HLCA’s finer taxonomy has *alveolar / elicited / lung macrophage* but **no generic macrophage** (CL:0000235), while Krasnow labels many cells *generic macrophage* — so an HLCA-trained model predicts *lung macrophage* (CL:1001603) for them, exact-wrong but ontology-correct (*lung macrophage is-a macrophage*); likewise Krasnow’s *generic endothelial cell* (CL:0000115) has no HLCA counterpart. **Ontology-aware concordance — which credits a same-lineage call — is ~0.75 for the trained methods (0.67–0.79 across the full panel, highest for ProtoCloud)**, so cross-dataset transfer actually works; exact-match just conflates classifier error with vocabulary/granularity mismatch. This is precisely why we report ontology concordance, and it is a general benchmarking pitfall: **cross-dataset exact accuracy between independently-annotated atlases is not a meaningful accuracy signal.** The other five datasets use a single shared label

vocabulary for reference and query (a split of one atlas, or HLiCA’s harmonized annotations), so their exact scores are unaffected.

3.2 The accuracy $\times$ speed frontier

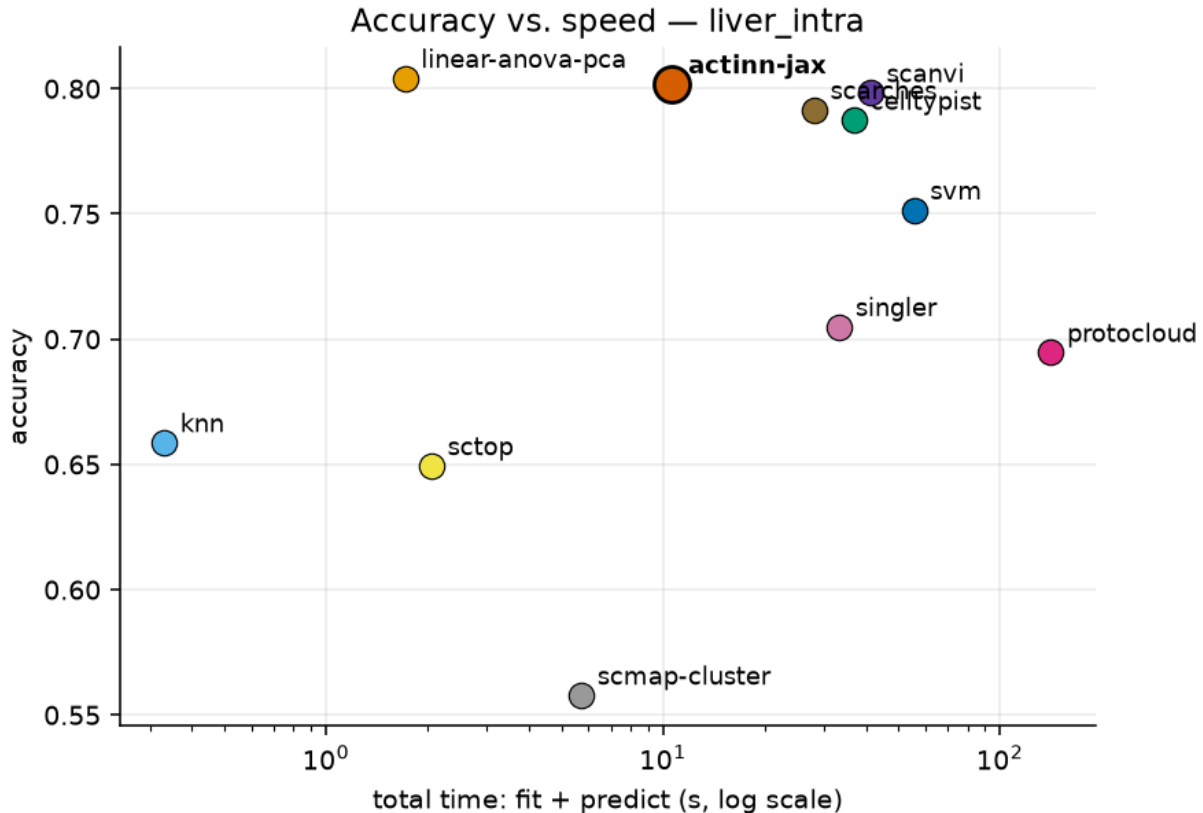


Figure 1: Accuracy versus total wall time (liver_intra); actinn-jax sits on the fast frontier.

Plotting accuracy against total wall time makes the frontier explicit. The **tuned linear pipeline** holds the fast-frontier point: highest matrix accuracy (0.839) at the lowest fit time of any accurate method (3.0 s), placing it above and to the left of actinn-jax, SVM, kNN and CellTypist. The deep methods buy little or nothing on accuracy here — at or below the linear pipeline on every dataset except lung — for 30–130 $\times$ the inference cost. actinn-jax sits just inside the frontier at this reference size; what separates it is not visible on a fixed-size plot but on the scaling axes of §3.3 and §5: predict time that stays flat as the reference grows, and $\sim 2\times$ lower peak memory that holds to atlas scale. The figure is drawn from the liver_intra panel and depicts the ranking in the tables above.

3.3 Scaling

Training time grows with reference size and with #cell types for all trained methods — actinn-jax’s fit goes 2.6 s $\rightarrow$ 17.4 s $\rightarrow$ 23.5 s as the reference grows 965 $\rightarrow$ 14.8k $\rightarrow$ 24k cells, below CellTypist (6 s $\rightarrow$ 71 s $\rightarrow$ 116 s) and SVM (4 s $\rightarrow$ 51 s $\rightarrow$ 80 s) at every size in the sweep; kNN is lazy (fit is trivial but it pays at predict and is the least accurate).

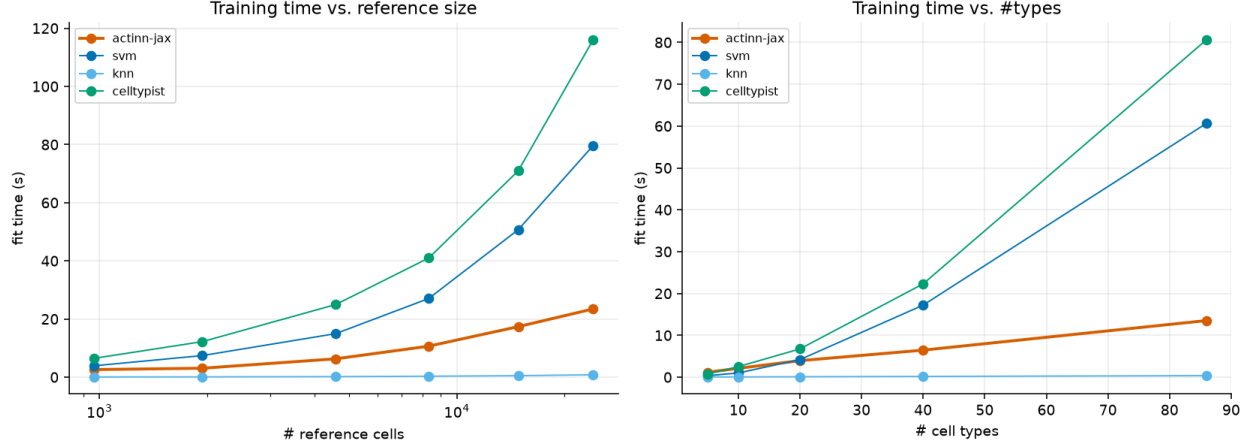


Figure 2: Fit and predict time versus reference size and cardinality; predict time stays flat and sub-second.

The other axis is the one that matters for the workflows: **predict time stays flat and sub-second across the entire range — 0.08–0.33 s whether the reference has 1k or 24k cells, or 5 or 86 types**. Inference cost does not grow with reference size or cardinality, and with the train-once/map-many cache the fit cost is paid once and the flat predict cost is all that recurs — the regime that matters when a reference is reused across many queries. (The scaling script’s memory column is process-cumulative and not a clean per-size measurement; per-method peak memory is reported from the isolated-subprocess matrix in §3.2.)

3.4 The broad→focused annotation workflow

What actinn-jax is built around is not a single classifier but a **two-pass workflow** that matches how annotation is actually done: get a broad call fast, then sharpen it where it matters. Because every stage is a cached **ReferenceModel** with sub-second, memory-bounded inference (§3.1, §3.3), the whole workflow runs on a laptop.

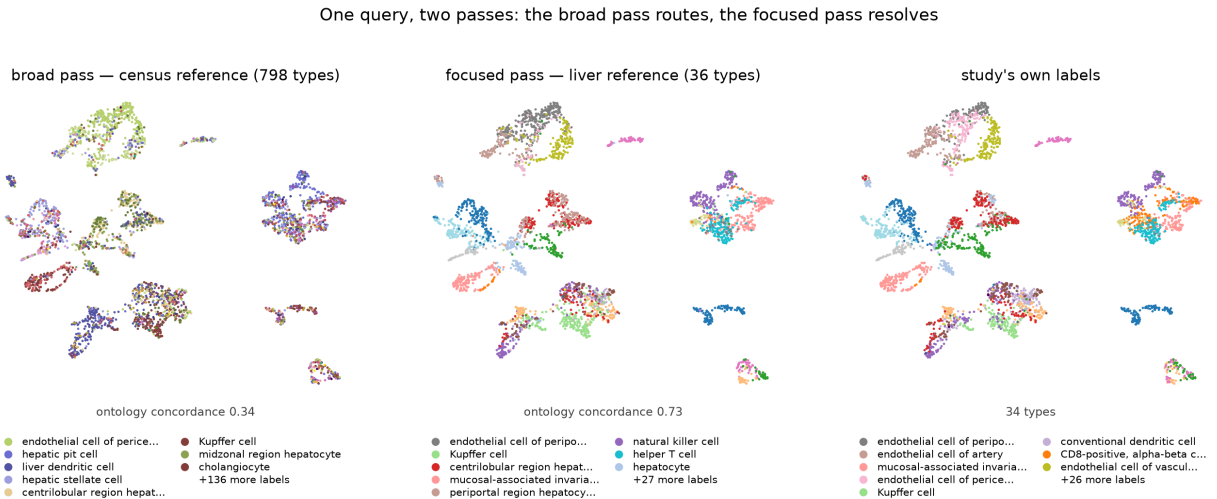


Figure 3: the broad pass and the focused pass on one query

Figure 5. The two passes on the withheld HLiCA liver study (3,396 cells), same UMAP throughout. *Left:* the shipped census reference spreads 137 of its 798 labels over the query with little correspondence to cluster structure — ontology concordance **0.34**. Its job is to establish that this is liver and route accordingly, not to name subtypes. *Middle:* a 36-type liver reference trained on the six non-withheld studies re-annotates the same cells at **0.73**, and its labels now track the clusters. *Right:* the study’s own labels, on the shared palette, for comparison with the middle panel. Both models are leakage-free with respect to this query: the focused reference is trained only on the other six studies (the *shipped liver_hlica_v2* includes this study and would score 0.94 here, which is why it is not the model plotted). Abstention is switched off in all panels so the numbers measure labelling rather than coverage; §3.5 covers abstain separately.

The broad pass — census-scale. A shipped ~800-type human reference built from the CEL-LxGENE census gives any query a first-pass annotation across the whole body, with a **calibrated abstain threshold** so out-of-reference cells are flagged rather than force-labeled (see the smooth abstain curve of §3.5). This is the “what am I looking at” pass — broad coverage, deliberately not fine-grained.

The broad pass can be built from a model instead of from labels. The census route above needs a foundation model on a GPU to discover its hierarchy. A pretrained pan-human annotator already has one — Pan-human Azimuth publishes an 8-level typology with every node mapped to a Cell Ontology term — so labelling a corpus with it and training actinn-jax on those labels transfers both the vocabulary and the structure. That build needs **no accelerator and no labeled input**, only raw human counts: under ten minutes of CPU on 85k cells drawn from three atlases plus a census-wide sample. On 3,396 withheld cross-study liver cells:

broad-pass model	classes	ontology	cells/s
census-built (broad_human_v1)	798	0.338	2,962
Pan-human Azimuth	382	0.380	1,076–1,563
distilled (panhuman_distill_v1)	324	0.406	8,937–10,021

This makes the entry point both better and ~3× faster than building it from the census directly, and it answers in a harmonized, ontology-mapped vocabulary rather than raw census strings. Withholding a whole atlas from the corpus, the distilled model tracks the one it was distilled from to within **1.5 points** on lung (0.695 vs 0.710) and **3.0** on liver (0.481 vs 0.511) — the corpus, not the recipe, is what bounds it. Two caveats keep this modest. The 0.406/0.380 ordering is one query, and both actinn-jax models draw on a census sample that may include these studies while Pan-human Azimuth does not, so read the distilled model as *comparable to* the one it distills rather than better. And it does **not** inherit that model’s trained abstention: its confidence separates right from wrong poorly (90.5% coverage at $p \geq 0.5$ moves concordance only 0.406 → 0.427), so the calibrated broad-pass abstain of §3.5 belongs to the census-built reference until this one is recalibrated.

A second organism, and a hierarchy with no accelerator in it. Neither route above transfers to mouse: Pan-human Azimuth is human-only, so there is no teacher to distill, and the census route wants scPRINT on a GPU with mouse support we have not tested. But the census already labels

every cell with a Cell Ontology term, and CL encodes the relation the embedding clustering is recovering — which cell types are kinds of the same thing. Describing each type by the CL terms it descends from and clustering *those* (same Ward linkage, ontology indicator vectors in place of embeddings) gives a hierarchy that is free, deterministic and species-independent.

That substitution needs a control, since any grouping might help. Built from one human corpus (51,346 cells / 867 types) and scored on the held-out lung atlas ([results_hierarchy_ablation.csv](#)):

coarse hierarchy	ontology
Cell Ontology lineage	0.616
random grouping, same group sizes	0.539
flat, no hierarchy	0.547

Random lands on flat, so it is the structure doing the work rather than the mere presence of groups. On that basis, **broad_mouse_v1**: 27,026 cells → **453 cell types across 85 tissues**, 305 CL terms collapsed to 21 coarse groups, **17 seconds of CPU** to train, 38 MB. Two mouse datasets were excluded from the reference entirely and used as the test set — 12,646 cells, 137 truth types, 41 tissues, none of it seen in training:

	ontology	coverage	cells/s
all cells	0.638	100%	9,712
min_prob = 0.5	0.718	71%	

Its abstain calibration is also better behaved than the human census model’s — 0.88 accuracy at 82% coverage, against 0.80 at 46% — because mouse census carries fewer near-duplicate subtypes than human’s ~800-way vocabulary. Read across organisms with care: these are different queries, so the mouse numbers sit in the same regime as the human ones rather than being comparable to them. Two limits are worth stating on the spot. The ablation above was run on **human** and applied to mouse; CL is species-neutral by construction, but nothing here shows it groups mouse types as well. And mouse census is shallow in *datasets* — 51 in total, one embryo atlas holding 11.4M of 18.4M cells — so tissue breadth is good while lab and protocol breadth is far below human’s 487 datasets.

The focused pass — tissue-specific. For the tissue the broad pass identifies, a small focused reference re-annotates at full resolution. The gain is large and it is the crux of the workflow: on held-out **cross-study** liver cells, the broad census model scores exact-CL **0.23** / ontology **0.58**, while a focused **48-type HLiCA liver** reference on the *same cells* reaches **0.72** / **0.86**. Refinement is where fine-grained accuracy comes from; the broad model’s job is to route to it, not to be right about subtypes itself.

The two tiers switch; they do not stack — we tested this and it is worth stating plainly, because the opposite is the natural assumption. On the leakage-free cross-study liver split, substituting the stronger **Pan-human Azimuth** for the broad pass lifts it (ontology 0.380 vs 0.338 for our own broad model) but changes nothing downstream. Using that broad call to *narrow* the focused pass’s classes — the zero-retrain masking actinn-jax ships — makes the result **worse**, 0.731 → 0.708: the broad call matches the true lineage on 85.8% of cells, and the 14% it misses cost more than the 86% it gets right can gain, since a wrong mask discards the correct class outright. A **perfect** coarse call

would add only **+2.8 points** (0.759). Once the focused pass covers the tissue, there is almost no accuracy left for a broad model to contribute; its value is choosing which focused reference to load and covering cells none of them claims (PAN_HUMAN_AZIMUTH.md).

Supporting mechanisms. (i) A **foundation-model-guided coarse→fine hierarchy**: a foundation model discovers a coarse→fine structure *offline*, the fast CPU model uses it at inference — beats a flat classifier on 3/4 datasets, matches an expert hierarchy, beats a random-grouping control, and never depends on the foundation model at prediction time. (ii) **Within-cell-type resolution**: the same machinery resolves hepatocyte zonation (portal→central) at ~0.99 within-one-zone, generalizing across donors and datasets — a third tier below the cell-type label. Together these make actinn-jax a *pipeline* (broad → tissue → subtype/state), not just a classifier, at classical-method speed throughout.

3.5 Rejection / abstain

Holding out 9 of 36 cell types entirely from the HLiCA liver reference (so 1,350 query cells are genuinely out-of-distribution), and sweeping a confidence threshold:

threshold	actinn-jax: acc(kept) / coverage / OOD-flagged	CellTypist: acc(kept) / coverage / OOD-flagged
0.0	0.885 / 1.00 / 0.00	0.873 / 1.00 / 0.00
0.5	0.919 / 0.93 / 0.30	0.936 / 0.68 / 0.68
0.7	0.946 / 0.84 / 0.52	0.936 / 0.68 / 0.68
0.9	0.969 / 0.66 / 0.73	0.937 / 0.68 / 0.68

actinn-jax’s probability gives a **smooth, tunable** abstain curve — accuracy on kept cells rises 0.885→0.969 as coverage falls 1.00→0.66 and OOD-flagging climbs 0.00→0.73 — so a user can dial the precision/coverage/novel-cell-detection trade-off. CellTypist’s probabilities are **saturated** (near 0 or 1): the threshold jumps to 68% OOD-flagged at 0.5 and then does not move across 0.5–0.9, giving essentially one operating point rather than a tunable curve. scmap-cluster offers a single native “unassigned” decision (no sweep). A tunable threshold is what makes the abstain step usable as a routing decision in the workflow of §3.4.

3.6 Foundation-model zero-shot (scPRINT)

scPRINT is run separately as a zero-shot predictor (no training on the reference), as a reference point for the “just use a foundation model” alternative. On the lung dataset it scored **exact accuracy 0.029 / ontology concordance 0.201, taking 62 s for 2,694 query cells** — against actinn-jax’s 0.894 in 0.23 s — so as a *label* predictor it is both slow (~280× actinn-jax) and weak. It is reported on lung alone because its fixed vocabulary cannot score the other datasets: the liver and blood+gut sets contain Cell Ontology terms absent from its label set, and it declines them rather than guessing. Its pretrained classifier head also carries a **fixed label vocabulary** and requires CL ids, so it cannot emit labels outside that set; this is an inherent property of a zero-shot label head, not a defect, but it means the foundation model’s raw predictions are not a drop-in annotator. This is exactly why our two-stage workflow uses scPRINT’s **embeddings** — its learned structure — to shape a small trained model, never its zero-shot labels: the foundation model is valuable, its raw label predictions are not.

3.7 External validation: Open Problems label_projection

Our in-house benchmark is neutral but self-run. For an **independent** check, we ran actinn-jax through the community-standard [Open Problems](#) label-projection task — their 6 datasets, splits, and metrics — and slotted it into their published v2.0.0 leaderboard.

Placement. On a benchmark we did not design and cannot tune, **actinn-jax places 3rd of 17 on mean accuracy (0.837), 1st among all methods that complete every dataset**, beats its PCA-space mlp sibling, and posts the best accuracy of any completing method on the hardest dataset (tabula_sapiens, 160 types: 0.394 vs mlp 0.342). It is upper-mid on macro-F1 and **not the top method overall** — scANVI+scArches surgery and xgboost [Chen & Guestrin 2016] score higher on the datasets they finish, though both **fail to complete tabula_sapiens**, which is why they rank above actinn-jax only on a mean over the 5 easier datasets. The foundation models again land at the bottom (scgpt_zeroshot 0.639, uce 0.131 $\approx$ the random-labels control), reproducing §3.6 externally.

Controlled same-hardware speed & memory. Leaderboard runtimes mix hardware, so we re-ran actinn-jax and the CPU tier through OP’s *own* Nextflow pipeline on a **single AWS r7i.8xlarge** — identical hardware, harness, storage, and trace instrumentation for every method. This is the paper’s cleanest cross-method cost comparison:

method (same box)	mean acc	macro-F1	runtime/dataset	peak RSS
actinn-jax	0.837	0.655	165 s	21.0 GB
mlp	0.838	0.664	327 s	19.9 GB
logistic_regression	0.813	0.689	29 s	20.0 GB
knn	0.793	0.648	11 s	19.5 GB
xgboost	0.795	0.613	905 s	80.7 GB
cellmapper_linear	0.776	0.553	58 s	31.5 GB
naive_bayes	0.738	0.613	31 s	19.5 GB

actinn-jax **ties its mlp sibling on accuracy at $\sim 2\times$ the speed** (165 vs 327 s) and **beats xgboost’s accuracy at $\sim 5.5\times$ less runtime and $\sim 4\times$ less memory** (165 s / 21 GB vs 905 s / 81 GB). The faster methods (knn, logistic) are less accurate; the more accurate heavyweight (xgboost) is far slower and heavier — Pareto-non-domination **within Open Problems’ method set**, on controlled hardware. That set contains no carefully-tuned linear pipeline, and where we add one ourselves it leads on accuracy and fit time (§3.1), so this is a statement about the OP panel rather than a general one.

Extension: our four added methods on the same harness. We later ported the tuned linear pipeline, scTOP, the SGD-SVM and CellTypist into Open Problems components and ran them through the same pipeline on a fresh r7i.8xlarge ([results_openproblems_samehw_v2.csv](#), raw trace archived alongside). All four run correctly on a harness we did not write, which is the main thing this establishes; peak memory sits in the same 13–20 GB band as the incumbent methods (xgboost is the outlier at 55 GB). **We do not report cross-method runtimes from this run, and the reader should not infer them from the CSV.** Two confounds make the wall-clock numbers non-comparable: the run used a higher task concurrency (6 vs 2), and the methods differ enormously in how many cores they take — measured %cpu ranges from **49% for CellTypist (half a core) and 117% for mlp (effectively single-threaded) to 2338% for the linear pipeline (23 cores)**. Wall-clock under those conditions measures threading and contention as

much as algorithmic cost. It also disagrees with the controlled run above for two methods (`mlp` 327 s $\rightarrow$ 769 s, `cellmapper_linear` 58 s $\rightarrow$ 717 s), which we flag rather than reconcile: the table above, run at low concurrency, remains the cost comparison we stand behind. Coverage was partial when a wall-clock budget stopped the run — six datasets for the SVM and CellTypist, four for scTOP, three for the linear pipeline — so per-method means over different dataset subsets would not be comparable either.

Three cheap ablations, and their limits. (i) *Input standardization* — z-scoring genes to the reference’s frozen mean/std, a CPU-only domain-alignment inspired by scArches reference surgery — lifts mean accuracy +0.2 and macro-F1 +1.2 pt (largest on batch-shifted datasets: gtex macro-F1 +7.9). It is shipped **opt-in** (`standardize=True`) because it shifts the softmax calibration that the two-stage abstain thresholds are tuned against. (ii) *Gene budget* — OP feeds every method 1000 HVGs, which most starves a gene-space MLP. Widening to ~5000 HVGs lifts accuracy on 4 of 6 datasets (immune/gtex +3 pt) and there **matches scANVI+scArches’s full-config leaderboard accuracy** (immune 0.891 $\approx$ 0.892) on CPU in minutes — the gap to the top method was largely a gene-budget artifact, not the VAE. But more genes is **not** a universal win: it *regresses* tabula_sapiens by ~10 pt (its 284-cell test batch across 160 fine types overfits reference-specific genes) and saturates hypomap. Crucially, this is **selectable without test labels**: held-out *reference* cross-validation rises for the datasets that benefit and is the one signal that *drops* for tabula_sapiens, and a trivial query-cells-per-class check independently flags it — so the budget can be set deterministically per dataset. (iii) *Negative control* — a CPU-only, UCE-style protein-embedding featurization (expression-weighted mean of ESM2 gene embeddings) does **not** help and hurts the hardest case: the pooling discards the per-gene detail that separates fine types, so the value of a foundation model like UCE stays locked in its GPU transformer, not a portable averaging trick.

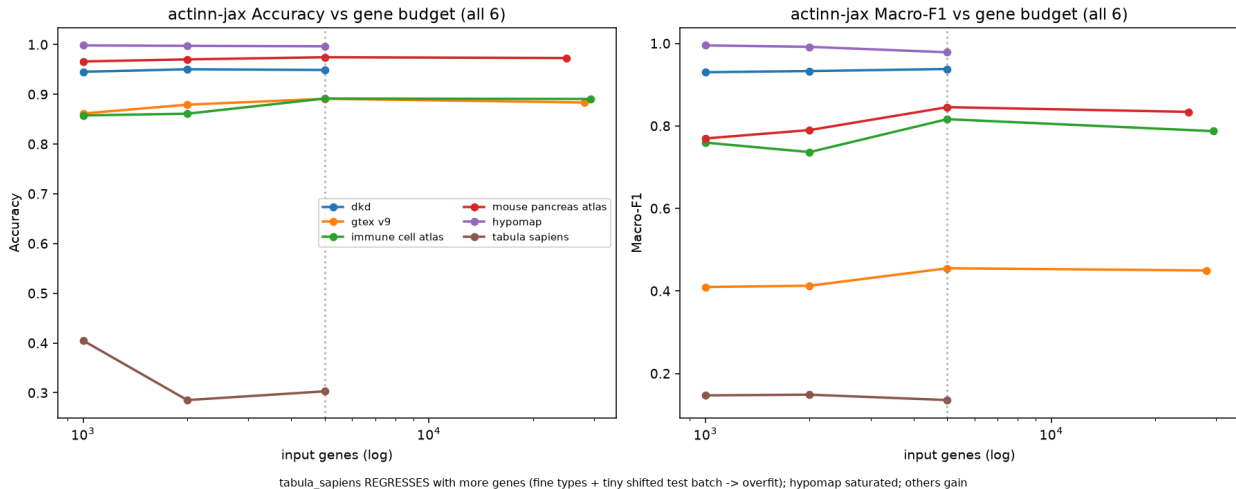


Figure 4: actinn-jax accuracy and macro-F1 versus input gene budget across all six Open Problems datasets. More genes help most datasets but regress the fine-grained, domain-shifted tabula_sapiens.

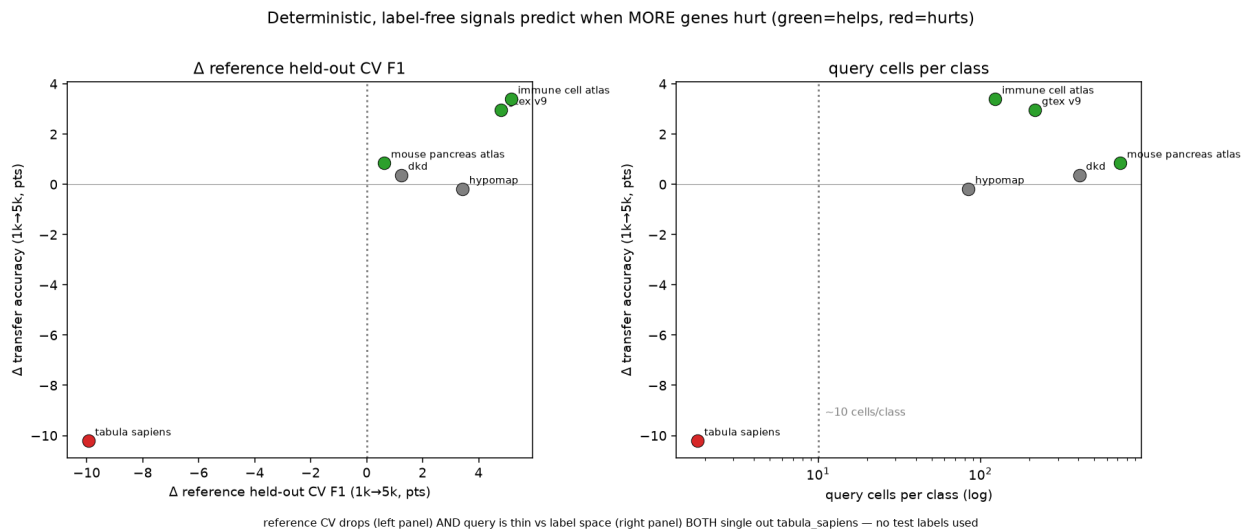


Figure 5: Label-free signals for setting the gene budget without test labels. Held-out reference cross-validation and query-cells-per-class both single out tabula_sapiens (where more genes hurt).

4 Discussion

Method choice is now dominated by cost, not accuracy. The five leading methods are separated by 0.008 in mean accuracy and by $\sim 130\times$ in inference time (§3.1), so for most annotation jobs the decision is made on the cost axes. On laptop-sized references the tuned linear pipeline is the best of them on accuracy-per-second (0.839, fitting in 4.0 s); at atlas scale ProtoCloud is the most accurate by a clear margin (0.976 vs 0.936 at 49k lung cells; 0.905 vs 0.824 at 47k liver cells), having been below the cluster on the subsampled matrix. actinn-jax’s place in that picture rests on a cost profile with two properties that matter for repeated use rather than for a single labelling run.

Footprint. Inference is **sub-second and flat** — independent of reference size and cardinality (§3.3) — so a query costs the same against a 1k-cell reference and a 49k-cell one. Peak memory is unremarkable on small references (2.1 GB mean — seven of the ten other methods are lighter), but it is the lowest of those we carried to **atlas scale**: $\sim 2\times$ below the linear pipeline (6.1 vs 13.2 GB at 49k lung cells; 6.5 vs 12.6 GB at 47k liver cells), and below scTOP above ~ 25 k cells, where scTOP’s rank processing densifies and crosses over (9.3 vs 6.5 GB at 47k). The advantage is bounded at $\sim 2\text{--}3\times$ rather than widening. Combined with the train-once/map-many cache, a reused reference pays fit once and then only the flat predict cost, where the linear pipeline refits scaler/PCA/classifier for every query.

Those two properties are what the workflows are built on. The practical gains come not from the flat classifier but from the **large→refined** pipeline (§3.4): a census-scale broad model with a calibrated abstain routes a query to a small focused reference that re-annotates at full resolution (cross-study liver 0.23/0.58 $\rightarrow$ 0.72/0.86), with zonation as a further tier. The pipeline is only usable because each hop is cheap — a flat sub-second call against a cached model, at bounded memory — so several models can be chained, and a large reference kept resident, on a laptop without a GPU. A method that is a few points more accurate but pays a full fit per query, or a multi-gigabyte dense expansion per stage, does not compose the same way.

Where it loses, and why. actinn-jax trails on small references and very-few-cells-per-type sets (§3.1, §5), and — a finding from the OP analysis — it is sensitive to input budget under domain shift: more genes help most datasets but overfit a tiny, fine-grained, shifted query (tabula_sapiens). The two cheapest levers are principled and label-free: **standardization** (a scArches-style domain alignment) and a **gene budget chosen by cross-validation on the reference alone** both improve the model without peeking at test labels, and at a fair gene budget the gene-space MLP *matches the leaderboard’s top method* on the datasets both complete — In other words, much of what looked like a deficit of the *model* was a deficit of its **input**: the benchmark hands every method 1,000 highly variable genes, which is a far harsher constraint on a model reading gene space directly than on one that learns a compressed representation first. Widen the gene panel and most of the gap closes — so the quantity being measured was partly “how well does each method tolerate a narrow input”, not “which model class is stronger”.

Foundation models. Their **zero-shot labels** are the weakest option in both benchmarks (scPRINT, scGPT, UCE $\approx$ random); their **embeddings/structure** are where the value lies, and our two-stage hierarchy uses exactly that. A cheap CPU shortcut to a foundation-model representation (protein-embedding pooling) did *not* transfer (§3.7), underscoring that the value is in the trained transformer, not a portable trick. The concurrent **Pan-human Azimuth** effort reaches the same conclusion from a far larger curated corpus: a 7M-parameter supervised hierarchical MLP over a 5,055-gene panel yields cleaner annotations than scGPT and SCimilarity, whose labels they find fragmented and mis-transferred. Their stated theme — that training-data quality and organization matter as much as architecture or scale — and their finding that accuracy **saturates past ~5M training cells** (PAN_HUMAN_AZIMUTH.md) are independent support, from a group with no stake in our conclusion, for both halves of our reading.

When extra expressivity pays. Two lines of evidence bear on this. First, **ProtoCloud** — a prototype-based, self-explaining VAE with built-in uncertainty and gene-level attribution, a strictly richer model than ours — does not sit a fixed distance above or below us. Where it lands depends on how much reference data it is given: below the top cluster on the subsampled matrix, ahead on lung, and the strongest method of all at atlas scale (§3.1, §5), at $\sim 9\times$ actinn-jax’s CPU fit time (146 vs 16 s mean) (PROTOCLOUD.md). The richer model pays off where it has the data to support it, and not before — which is an argument for matching model capacity to reference size, not for preferring small models everywhere. Second, and more broadly, Souza & Mehta [Souza & Mehta 2026] report that **parameter-free linear representations** match or exceed single-cell foundation models across cross-species transfer, human cell-type classification and disease-state prediction — on Tabula Sapiens 2.0 their normalize→ANOVA→PCA→logistic-regression pipeline reaches mean macro-F1 0.899 against 0.907–0.910 for TranscriptFormer variants — and they offer a geometric explanation: the biologically realized cell manifold is well approximated by a **linear subspace**, so once noise is suppressed performance *saturates* and extra expressivity buys little. That is a principled account of why a four-layer MLP over normalized gene space stays competitive, and it matches our own finding that the residual gap to heavier methods was mostly **the size of the input each method was given rather than the class of model** (§3.7). They further find that simple methods hold up *best* out-of-distribution — on novel cell types and unseen organisms. That matters for the abstain mechanism of §3.5, which is only useful if low confidence tracks genuine novelty rather than model noise: a method whose out-of-distribution behaviour degrades gracefully is one whose confidence can be thresholded.

Practical guidance. For a **one-off annotation** on laptop-sized data, use the **tuned linear pipeline** (normalize → ANOVA → PCA → logistic regression): it is the most accurate-per-second

method here. With a **real atlas and time to train**, **ProtoCloud** is the most accurate method we benchmarked (0.976 at 49k cells), and the GPU it targets makes its 19× CPU fit cost much less punishing in practice. **scTOP** suits small, low-cardinality problems, where it is by far the cheapest (~1 s) and competitive. **actinn-jax** fits the cases its cost profile is shaped for: when **memory is the binding constraint** (~2× lighter than the linear pipeline at scale, with memory-bounded chunked inference), when a reference is **reused** across many queries (the cached model amortizes the fit), and when the job is the **multi-stage workflow** rather than a single label — broad→refined hand-off, tissue-aware refinement, calibrated abstain, novel-cell-type screening.

For the **broad tier specifically**, prefer a **purpose-built pan-human model**: **Pan-human Azimuth** (PAN_HUMAN_AZIMUTH.md) ships a pretrained hierarchical classifier over a harmonized organism-wide typology (8 levels, 382 leaf types, trained on 9.7M curated cells), with abstention learned rather than thresholded and an **expected calibration error (ECE)** of 0.0044 — meaning its stated confidence sits within half a percentage point of its observed accuracy — running at ~1,000 cells/s on a laptop. It is better resourced than our census-built reference and we make no claim to improve on its annotations. What we do with it instead is **use it**: distilling its labels and its hierarchy into actinn-jax produces a broad-pass model of comparable accuracy at roughly nine times its throughput, built without a GPU and without labeled data (§3.4). A curated pan-human model is the right thing to start from; a small fast model is the right thing to iterate with.

The part of any fixed typology that cannot be performed is the **hand-off**: it cannot re-annotate into a user’s own focused label set (the HLiCA liver reference, cross-study 0.23/0.58 → 0.72/0.86) or resolve states below a leaf (hepatocyte zonation). ProtoCloud likewise provides uncertainty, gene attribution and a refinement path (fine-tuning a pretrained model). What survives as distinct to actinn-jax is therefore not the breadth of the broad reference but **the chain it enables** — start pan-human, re-annotate into label sets no pretrained model carries, screen what none of them claims — at a cost that keeps every stage on a laptop. The distillation recipe is not specific to this teacher or to humans: it needs a pretrained annotator whose typology can be read off its outputs, so extending the same entry point to other organisms is a matter of finding one (or building the broad-pass reference from a labeled census slice, as §3.4’s census route does).

What the speed is for. Annotation is almost never the result; it is the step before the biology, and it is usually run more than once — the labels change when a reference is swapped, a threshold moves, a new sample arrives, or a cluster turns out to be two things. The argument of this paper is not that a faster classifier produces better labels; at equal data most of these methods produce much the same labels (§3.1). It is that when a stage costs a second instead of a minute, and a chain of stages fits in memory on the machine already on the desk, the loop from question to annotated data to the next question closes in an afternoon rather than across a queue. That is the contribution we would defend: not a better number in a table, but a workflow — start from the best pan-human model available, refine into the labels a given study actually uses, and flag what neither covers — that a working scientist can run, inspect, and run again.

5 Limitations

- Single hardware family for the in-house panel (Apple Silicon); no discrete-GPU numbers there (deep/foundation tiers would be faster on CUDA — out of scope for the “runs-on-a-laptop” question). The §3.7 controlled run covers the **CPU tier** on one cloud box; the GPU/R methods there are reported from OP’s own cloud-CI trace (indicative). A same-hardware GPU-tier run to fold those into a single controlled table is the natural extension.

- **Wall-clock comparisons across methods are only meaningful at matched concurrency, and we learned this the expensive way.** A follow-up run that added our four components to the OP harness (§3.7) used higher task concurrency and produced runtimes that disagree with the controlled run by up to 12× for the same method. Measured %cpu across methods spans 49% to 2338% — some are single-threaded, some saturate 23 cores — so any wall-clock ranking silently ranks threading and scheduler contention alongside algorithm. The controlled low-concurrency run is the one we report; the extension establishes only that the components run and what memory they need.
- Subsampled references (to keep the matrix tractable); the scaling section characterizes the size dependence directly.
- **The 12-method comparison is human only;** six datasets per benchmark; GPU foundation models beyond scPRINT/UCE (scGPT, Geneformer, popV) not run locally. The shipped references now cover mouse (§3.4), but no *method comparison* was run on mouse data — the pan-mouse result establishes that the reference-building route works on a second organism, not that actinn-jax’s standing against other methods carries over to it.
- actinn-jax needs more cells/type than linear methods to reach parity (§3.1); on very small references it trails.
- **The distilled broad-pass reference inherits a vocabulary, not a calibration.** Pan-human Azimuth’s abstention is trained (and its ECE measured at 0.0044); the distilled student learns hard labels only, so it carries none of that — its confidence barely separates right from wrong (§3.4). Recalibrating it, or distilling from soft targets, is the obvious next step and would need a loss change in the package. Its evaluation is also two withheld *atlases*, not withheld tissue: the census-wide corpus spans 376 tissues, so nothing here measures behaviour on biology the corpus never saw.
- **The gene budget is dataset-dependent, not a free parameter.** More genes help most references but *overfit* a tiny, fine-grained, domain-shifted query (tabula_sapiens −10 pt); our reference-cross-validation + cells-per-class selection rule is validated on 6 datasets with a single clean failure case, so it is directional evidence, not a tuned threshold.
- **Standardization is shipped opt-in,** not default, because it shifts probability calibration that the two-stage abstain thresholds are tuned against; combining the two cleanly would require re-tuning those thresholds.
- **Our classical tier (SVM, kNN, CellTypist) is untuned, and a tuned linear baseline beats actinn-jax.** Souza & Mehta’s pipeline (normalize → ANOVA → standardize → PCA(220) → logistic regression) leads the matrix on accuracy (0.839 vs 0.831) and macro-F1 (0.699 vs 0.683) while fitting 4× faster, with its largest margin on the 86-type blood+gut set (+4.2 pt) (SIMPLE_BASELINES.md). The §3.1 margins over the classical tier should therefore be read as margins over *untuned* baselines; a like-for-like tuning effort on SVM or CellTypist would likely narrow them further. scTOP is cheaper still on time (~1 s) but competitive only at small, low cardinality — best pbmc macro-F1 (0.837), degrading to 0.649 on liver.
- **actinn-jax’s memory advantage is bounded, and the linear pipeline scales further than extrapolation suggests** (SCALING_MEMORY.md). Extrapolating the linear pipeline’s small-reference footprint predicts ~32 GB and failure at atlas scale; measured, it uses **13.2 GB** and completes, because SelectKBest(k=20000) caps the dense matrix well below the full gene set. The linear/actinn-jax memory ratio is likewise **bounded at ~2–3×** (2.15× at full scale) rather than widening with data, so there is no scaling cliff and no regime tested where the linear pipeline stops being usable. Memory projections in this setting should be measured, not extrapolated.
- **At atlas scale the ordering changes and the deep model leads.** Scaling the lung

reference 3k $\rightarrow$ 49k cells, **ProtoCloud** goes from worst (**0.722**) to best (**0.976**), clear of actinn-jax (0.936) and the linear pipeline (0.939), at $19\times$ the CPU fit cost; scTOP gains nothing from scale (0.819 $\rightarrow$ 0.846) and loses its memory edge ($1.22\times$ actinn-jax at 49k). This replicates on the HLiCA liver atlas (36 types, 525k cells, sweep to 47k ref): same bounded $\sim 2\times$ memory band, same ProtoCloud reversal (0.581 $\rightarrow$ 0.907), same scTOP stagnation (0.657 $\rightarrow$ 0.583), with scTOP crossing over to heavier than actinn-jax ($1.41\times$). Conclusions drawn from subsampled references do not transfer to atlas scale, in either direction.

- **Annotation only.** Cross-species transfer and disease-state prediction — the tasks where Souza & Mehta find the largest foundation-model deficits, and where a fast method would be most attractive — are outside this benchmark’s scope (human, within/cross-dataset annotation).

Data & code availability

All configs (`configs/paper*.yaml`, including `configs/paper_baselines.yaml` which fills the matrix for the three later-added methods on the identical splits), adapters (`benchmark/adapters/`), result CSVs (`results/paper*/`, `docs/results_*.csv`, unified matrix in `docs/results_paper_matrix_unified.csv`), figure scripts, and the actinn-jax package (github.com/iandriver/actinn-jax) are in these two repositories. Rebuilding the shipped broad reference is a single documented command (`benchmark/explore/update_broad_reference.sh`); the distilled broad-pass reference is built by `distill_dump.py  $\rightarrow$  distill_train.py`; the pan-mouse reference and the Cell-Ontology hierarchy it depends on are in .

Pre-trained references — human (`broad_human_v1`, `panhuman_distill_v1`), mouse (`broad_mouse_v1`) and focused liver (`liver_hlica_v1/v2`) — are archived at [doi:10.5281/zenodo.21688151](https://doi.org/10.5281/zenodo.21688151) (CC BY 4.0; cite the concept DOI [10.5281/zenodo.21688150](https://doi.org/10.5281/zenodo.21688150) for the latest version). `actinn_jax.bundled_references` downloads and caches them on first use, so no manual retrieval is needed; `actinn-jax fetch` pre-downloads for offline use.

HLiCA data © Edgar et al. 2026 (CC-BY 4.0), [[doi:10.64898/2026.06.30.735539](https://doi.org/10.64898/2026.06.30.735539)]. The distilled reference derives from **Pan-human Azimuth** (Sarkar, Li, Molla, ... Satija, bioRxiv 2026, [doi:10.64898/2026.07.16.738997](https://doi.org/10.64898/2026.07.16.738997)); its weights are © the authors under [CC BY 4.0](https://creativecommons.org/licenses/by/4.0/), obtained via `panhumanpy` (MIT) and [Zenodo](https://zenodo.org/). Attribution is a condition of that licence and travels inside the shipped model’s `build_info.json`.

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